Project Report

AI-Based Daily Routine Analyzer

---

1. Introduction

In today's fast-paced lifestyle, maintaining a healthy and balanced daily routine has become challenging. Students and working individuals often struggle with irregular sleep, excessive screen time, poor eating habits, and lack of physical activity. These habits can negatively affect productivity, health, and overall well-being.This project aims to analyze daily routine habits using machine learning and classify them into meaningful categories. The system evaluates user inputs and provides feedback to improve lifestyle quality.

---

2. Problem Statement

Many individuals are unaware of whether their daily routine is healthy or not. There is a lack of simple tools that can evaluate daily habits and provide meaningful feedback.

The problem is to develop a system that can:

- **Analyze daily lifestyle habits**
- **Classify routine quality**
- **Provide suggestions for improvement**

---

3. Objective

The main objectives of this project are:

- **To collect and analyze daily routine data**
- **To build a machine learning model for classification**
- **To classify routines as Good, Average, or Poor**
- **To provide suggestions for improving lifestyle habits**
- **To apply AI/ML concepts to solve a real-world problem**

---

4. Methodology

The project follows a structured machine learning approach:

Step 1: Data Collection

A dataset was manually created based on common daily habits such as sleep hours, water intake, exercise, screen time, study hours, meals, and wake-up patterns.

Step 2: Data Preparation

The dataset was converted into a structured format using Pandas DataFrame. The data was divided into:

- **Features (input variables)**
- **Target (output label)**

Step 3: Model Selection

A Decision Tree Classifier was selected because:

- **It is simple and easy to understand**
- **Works well with small datasets**
- **Suitable for classification problems**

Step 4: Model Training

The model was trained using the dataset. It learned patterns between lifestyle habits and routine quality.

Step 5: User Input

The system takes input from the user regarding their daily routine.

Step 6: Prediction

The trained model predicts whether the routine is Good, Average, or Poor.

Step 7: Suggestions

Based on user input, the system provides recommendations to improve daily habits.

---

5. Algorithm Used

Decision Tree Classifier

A Decision Tree is a supervised machine learning algorithm used for classification and regression tasks. It works by splitting data into branches based on conditions.

In this project, the Decision Tree learns rules such as:

- **Low sleep and high screen time → Poor routine**
- **Balanced habits → Good routine**

The model makes decisions based on feature values and predicts the output category.

---

6. Implementation

The project is implemented using Python and the following libraries:

- **Pandas for data handling**
- **Scikit-learn for machine learning**

The system:

- **Creates a dataset**
- **Trains a Decision Tree model**
- **Takes user input**
- **Validates input values**
- **Predicts routine category**
- **Displays suggestions**

---

7. Results

The model successfully classifies daily routines into:

- **Good**
- **Average**
- **Poor**

The system provides meaningful suggestions such as:

- **Improving sleep duration**

- **Reducing screen time**
- **Increasing physical activity**
- **Maintaining proper meals**

The output is simple, understandable, and useful for users.

---

8. Key Decisions

- **Using Decision Tree for simplicity and interpretability**
- **Creating a custom dataset based on realistic habits**
- **Adding input validation for better usability**
- **Including suggestion logic for practical value**

---

9. Challenges Faced

- **Designing a meaningful dataset manually**
- **Ensuring the model learns correct patterns**
- **Handling user input errors**
- **Balancing simplicity with functionality**

---

10. What I Learned

- **Basics of supervised machine learning**
- **Working with datasets using Pandas**
- **Training and using a Decision Tree model**
- **How to apply AI concepts to real-world problems**

---

12. Conclusion

This project demonstrates how machine learning can be used to analyze daily routines and provide meaningful insights. The Decision Tree model successfully classifies routines and helps users understand their lifestyle habits.

The system is simple, effective, and shows the practical application of AI/ML concepts in solving real-world problems.

Aryan Arora

25BAS10008

Vit Bhopal University